

BENCHMARK: INACCURATE

Results

Mid-Transit Time (Tmid)	2458022.7398 +/- 0.002 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.1568 +/- 0.0058
Transit depth (Rp/Rs)^2	2.46 +/- 0.18 [%]
Orbital Inclination (inc)	89.98 +/- 0.5
Airmass coefficient 1 (a1)	1.0572 +/- 0.007
Airmass coefficient 2 (a2)	-0.054 +/- 0.076
Scatter in the residuals of the lightcurve fit is	0.67 %
Best Comparison Star	#2 - [355, 41]
Optimal Aperture	4.38
Optimal Annulus	14.46
Transit Duration (day)	0.0851 +/- 0.002

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R $\blacksquare$ fit vs published	0.1568 $\pm$ 0.0058 vs 0.1592 (deviation 0.41 σ)
Duration fit vs published	0.0851 d vs 0.0946 d (deviation 4.74 σ)
Mid-time O-C	-1.4 min
Depth SNR	27.0
Residual scatter	0.67 %

Light curve

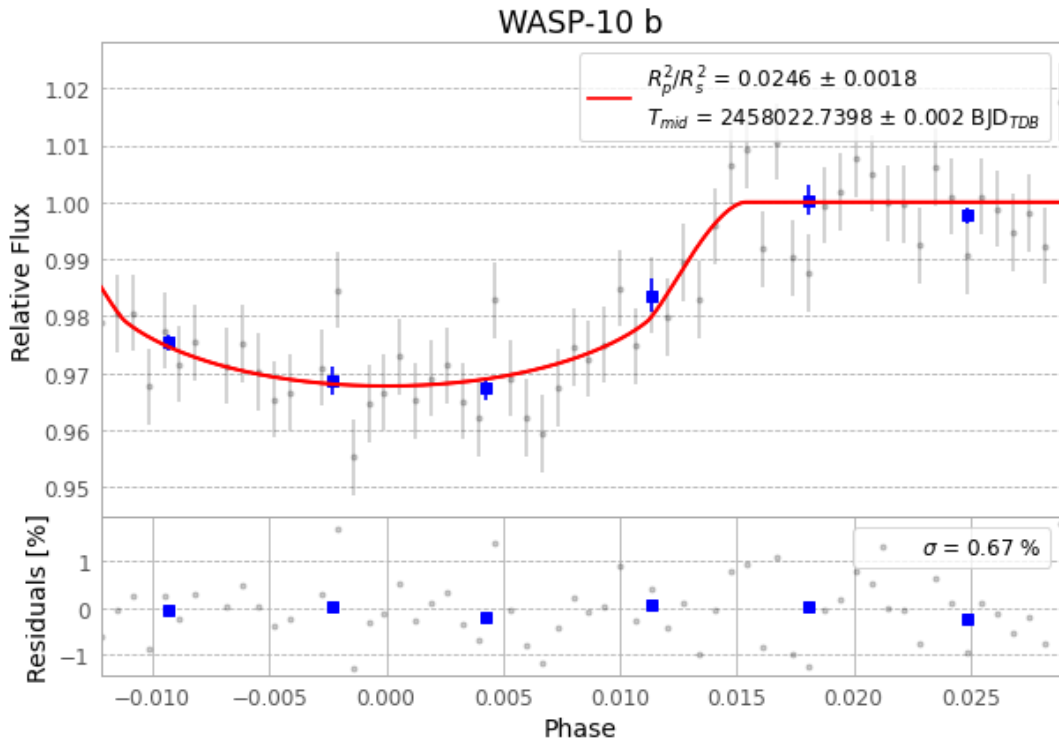


Figure 1 — FinalLightCurve_WASP-10 b_2017-09-25.png (SHA-256 1217f01708d66fcb...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [359, 141], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 9133a19e161b76c5...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ead60b9

Data provenance

62 FITS frames (41 MB), WASP-10170926044212.FITS → WASP-10170926074513.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-10-170926.log (72e2db9feb08...), FinalLightCurve_WASP-10 b_2017-09-25.png (1217f01708d6...), FinalLightCurve_WASP-10 b_2017-09-25.csv (e5d91d4ac199...)

Compute resources

Wall time 135.2 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-28 23:47Z.